Extremely projective normed spaces

A characterization in all dimensions

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Helemskii's seminar

Plan

1. Lifting problems
2. Basic theory
3. Dimension restrictions
4. Countable dimension
5. Extremely projective spaces which are not free
6. Uncountable dimension
7. Dimension $\aleph_1$
8. Summary

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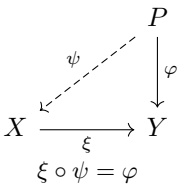
8. Summary

Lifting problems

Given a "surjection" $\xi\colon X \to Y$ and a map $\varphi\colon P \to Y$: does φ lift to X?

P is projective if every such problem has a solution ψ with $\|\psi\| \le cC\|\varphi\|$.

Which surjections? In $\mathbf{Nor}$, the category of *normed* spaces and bounded operators, the right notion is quantitative.



Definition

$\xi\colon X \to Y$ is c-topologically surjective ($c \ge 1$) if $c\,\xi(B_X^\circ) \supset B_Y^\circ$, i.e.

$$\forall y \in Y \quad \forall \delta > 0 \quad \exists x \in X : \quad \xi(x) = y, \qquad \|x\| < c\|y\| + \delta.$$

Coisometric: contractive and 1-topologically surjective, $\xi(B_X^\circ) = B_Y^\circ$.

Topological and extreme projectivity

Definition

A normed space P is C-topologically projective ($C \geq 1$) if for every $c \geq 1$, every c-topologically surjective $\xi\colon X \to Y$ between normed spaces and every bounded $\varphi\colon P \to Y$ there is a bounded $\psi\colon P \to X$ with

$$\xi\psi = \varphi, \qquad \|\psi\| \leq cC\|\varphi\|.$$

P is topologically projective if this holds for some C, and extremely projective if it holds for *every* $C > 1$.

- X and Y are arbitrary *normed* spaces. Restricting to Banach spaces gives the (different!) theory in $\mathbf{Ban}$.
- Normed spaces = normed modules over the zero algebra. This is the simplest case of Helemskii's *extreme projectivity* for normed modules; his definition looks different but agrees with this one (later).
- Extreme projectivity sits between the metric theory ($C = 1$) and the topological theory (some C).

Banach spaces: the classical answers

Theorem (Köthe 1966)

A Banach space is topologically projective in $\mathbf{Ban}$ $\iff$ *it is topologically isomorphic to* $\ell_1(\Lambda)$ *for some set* Λ.

Theorem (Grothendieck 1955)

A Banach space has the metric lifting property $\iff$ *it is isometrically isomorphic to some* $\ell_1(\Lambda)$.

Both proofs use completeness: open mapping, Pełczyński's decomposition method.

In $\mathbf{Nor}$ the free objects are different

$$\ell_1^0(\Lambda) = \{\text{finitely supported families on } \Lambda\}, \qquad \|x\| = \sum_\lambda |x_\lambda|.$$

Bounded families $(x_\lambda)_{\lambda\in\Lambda}$ in a *normed* space $E \leftrightarrow$ bounded operators $T\colon \ell_1^0(\Lambda) \to E$, $T(e_\lambda) = x_\lambda$, $\|T\| = \sup_\lambda \|x_\lambda\|$. No completeness of E is needed, in contrast with $\ell_1(\Lambda)$.

Normed spaces: what was known

Theorem (Grønbæk 2016)

E is topologically projective in $\mathbf{Nor}$ $\iff$ *E is topologically isomorphic to* $\ell_1^0(\Lambda)$, $|\Lambda| = \operatorname{dens}(E)$.

Theorem (Helemskii 2013)

E is metrically projective in $\mathbf{Nor}$ $\iff$ *E is isometrically isomorphic to some* $\ell_1^0(\Lambda)$.

Extreme projectivity (Helemskii 2013): $\ell_1^0(\Lambda)$ is extremely projective; P extremely projective $\iff$ P is a near-retract of some $\ell_1^0(\Lambda)$; P extremely projective $\Rightarrow \overline{P} \cong \ell_1(\Lambda)$ isometrically, so P is a dense subspace of some $\ell_1(\Lambda)$; P separable and topologically projective $\Rightarrow \dim P \leq \aleph_0$.

Question (Helemskii 2013; Grønbæk 2016, Conjecture 3.1)

Is every extremely projective normed space isometrically isomorphic to some $\ell_1^0(\Lambda)$? More precisely: *which dense subspaces of $\ell_1(\Lambda)$ are extremely projective?*

The answers, I

Theorem A (counterexample)

$\ell_1^0(\mathbb{N}) + \mathbb{C}v$, $v = (2^{-n})_n$, is extremely projective but not isometrically isomorphic to any $\ell_1^0(\Lambda)$.

Theorem B (almost isometric characterization)

For a normed space P the following are equivalent:

(i) P is extremely projective;

(ii) P is topologically projective and $\overline{P} \cong \ell_1(\Gamma)$ isometrically;

(iii) $d_{BM}(P, \ell_1^0(\Gamma)) = 1$ for some Γ;

(iv) for every $\varepsilon > 0$ there is Λ with $d_{BM}(P, \ell_1^0(\Lambda)) < 1 + \varepsilon$.

For $\dim P \le \aleph_0$: P extremely projective $\iff$ $\overline{P} \cong \ell_1$ or ℓ_1^n isometrically.

So Grønbæk's conjecture fails isometrically and holds almost isometrically.

The answers, II

Theorem C (dimension restrictions)

Topologically projective normed spaces contain no infinite dimensional complete subspaces. In particular, no infinite dimensional Banach space is topologically projective in $\mathbf{Nor}$.

Theorem D (dense subspaces of $\ell_1(\Gamma)$)

A dense $P \subset \ell_1(\Gamma)$ is extremely projective $\iff$ Γ has a continuous filtration with countable blocks along which the pieces are dense, form a continuous chain and have countable successive quotients.

Theorem E (dimension $\aleph_1$)

For $\Gamma = \omega_1$: $\iff$ the pieces $P \cap \ell_1(\delta)$ have countable dimension and the chain jumps only on a non-stationary set. There are $2^{\aleph_1}$ pairwise non-isomorphic dense subspaces of $\ell_1(\omega_1)$ with extremely projective countable pieces, none of them projective.

So for dense subspaces of $\ell_1(\Gamma)$ topological and extreme projectivity coincide, and for uncountable Γ this is a genuine restriction beyond Theorem C.

Why "normed" is the difficulty

Theorem C says: an extremely projective space never contains an infinite dimensional Banach space. So the objects are *incomplete in every direction*, and the usual toolbox is switched off:

- no Baire category, no open mapping, no closed graph;
- no Pełczyński decomposition method (ℓ_1-sums of the pieces are not inside P);
- no Grothendieck-type duality for P itself, only for $\overline{P}$.

What survives

The finite dimensional pieces of P, the Hamel basis, Hahn–Banach, the Neumann series in the *completion*, and the exact ℓ_1-norm.

The main tool will be a back-and-forth construction: an automorphism of $\ell_1(\Gamma)$ of distortion $< 1+\varepsilon$ built step by step, which moves P exactly onto $\ell_1^0(\Gamma)$. In the uncountable case the steps are indexed by ordinals and controlled on disjoint blocks of coordinates.

Plan

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Helemskii's definition is the same

Helemskii: P is *extremely projective* if for every *coisometric* $\xi\colon X \to Y$, every $\varphi\colon P \to Y$ and every $\delta > 0$ there is a lifting ψ with $\|\psi\| < \|\varphi\| + \delta$.

Proposition

Extremely projective in the sense of the definition above $\iff$ *extremely projective in the sense of Helemskii.*

Proof sketch.

($\Rightarrow$) Coisometric operators are 1-topologically surjective; take $C = 1 + \delta/(2\|\varphi\|)$.
($\Leftarrow$) Given a c-topologically surjective ξ, renorm X:

$$\|x\|' = \max\bigl(\|x\|,\ c\|\xi(x)\|\bigr).$$

Then $\xi' = c\xi\colon X' \to Y$ is coisometric: contractive by construction, and $y \in B_Y^\circ$ has a preimage $x \in B_X^\circ$ under $c\xi$ with $\|x\|' = \max(\|x\|, \|y\|) < 1$. A lifting ψ' along ξ' with $\|\psi'\|' < C\|\varphi\|$ gives $\psi = c\psi'$ with $\xi\psi = \varphi$ and $\|\psi\| \le c\|\psi'\|' \le cC\|\varphi\|$. □

Free spaces are extremely projective

Proposition

$\ell_1^0(\Lambda)$ is $(1+\varepsilon)$-topologically projective for every $\varepsilon > 0$.

Proof.

Let $\xi\colon X \to Y$ be c-topologically surjective and $\varphi\colon \ell_1^0(\Lambda) \to Y$, $\varphi \neq 0$. For each λ choose $x_\lambda \in X$ with

$$\xi(x_\lambda) = \varphi(e_\lambda), \qquad \|x_\lambda\| < c\|\varphi(e_\lambda)\| + c\varepsilon\|\varphi\| \leq c(1+\varepsilon)\|\varphi\|.$$

The family is bounded, so $\psi(e_\lambda) := x_\lambda$ defines a bounded operator with $\|\psi\| = \sup_\lambda \|x_\lambda\| \leq c(1+\varepsilon)\|\varphi\|$ and $\xi\psi = \varphi$ on the Hamel basis. □

The constant cannot be 1

The only 1-topologically projective normed space is $\{0\}$: a coisometric functional on $(c_{00}, \|\cdot\|_\infty)$ need not attain its norm (backup slide). So in **Nor** one considers all $C > 1$ at once.

Retracts and the canonical morphism

$\zeta\colon E \to F$ is a c-retraction if $\zeta\eta = 1_F$ for some η with $\|\zeta\|\|\eta\| \le c$; then F is a c-*retract* of E.

Proposition (permanence)

P' C-topologically projective, P a c-retract of $P' \Rightarrow P$ is cC-topologically projective.
Hence: a $(1+\varepsilon)$-retract of an extremely projective space for every $\varepsilon > 0$ is extremely projective.

For a normed space P consider the canonical morphism

$$\pi_P\colon \ell_1^0(B_P) \to P, \qquad e_x \mapsto x.$$

Contractive, and $\pi_P(B_{\ell_1^0(B_P)}) = B_P$: strictly coisometric.

Proposition

(i) *P C-topologically projective $\Rightarrow$ π_P has a right inverse σ with $\|\sigma\| \le C$ (lift 1_P along π_P);*
(ii) *π_P has a right inverse of norm $\le C \Rightarrow P$ is $C(1+\varepsilon)$-topologically projective for all $\varepsilon > 0$.*

One test case suffices: the universal surjection π_P.

The criterion

Theorem

For a normed space P the following are equivalent:

(i) *P is extremely projective;*

(ii) *for every $\varepsilon > 0$, π_P has a right inverse of norm $\leq 1+\varepsilon$;*

(iii) *for every $\varepsilon > 0$, P is a $(1+\varepsilon)$-retract of some $\ell_1^0(\Lambda)$;*

(iv) *for every $\varepsilon > 0$ there are Λ, a subspace $Z \subset \ell_1^0(\Lambda)$ and a projection Q of $\ell_1^0(\Lambda)$ onto Z with $\|Q\| \leq 1+\varepsilon$ and $d_{BM}(P, Z) \leq 1+\varepsilon$.*

$(i) \Leftrightarrow (iii)$ is Helemskii's near-retract criterion. $(iii) \Rightarrow (iv)$: normalize $\|\zeta\| = 1$, $\|\eta\| \leq 1+\varepsilon$, put $Z = \eta(P)$, $Q = \eta\zeta$. $(iv) \Rightarrow (i)$: $\zeta = T^{-1}Q$, $\eta = T$ make P a $(1+\varepsilon)^2$-retract of $\ell_1^0(\Lambda)$; use permanence.

Takeaway

For every ε, P sits inside a free space $(1+\varepsilon)$-isomorphically and $(1+\varepsilon)$-complemented. We shall construct **isomorphisms onto free spaces** of distortion $< 1+\varepsilon$.

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5. Extremely projective spaces which are not free

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Free spaces are small

For infinite Λ: $\dim \ell_1^0(\Lambda) = |\Lambda| = \operatorname{dens} \ell_1^0(\Lambda)$. An infinite dimensional Banach space has uncountable Hamel dimension (Baire), in fact $\geq \mathfrak{c}$ (Löwig 1934).

Theorem (hereditary dimension bound)

Let P be topologically projective and $Z \subset P$ a subspace. Then

$$\dim Z \leq \max(\operatorname{dens} Z, \aleph_0).$$

In particular every separable subspace of P has countable dimension.

Proof.

Take $\sigma\colon P \to \ell_1^0(B_P)$ with $\pi_P \sigma = 1_P$; σ is injective. Let $M \subset Z$ be dense, $|M| = \operatorname{dens} Z$, and $S_0 = \bigcup_{x \in M} \operatorname{supp} \sigma(x)$, so $|S_0| \leq \max(\operatorname{dens} Z, \aleph_0)$. For $z \in Z$, $\sigma(z)$ is a limit of elements supported in S_0; coordinate functionals are continuous, so $\sigma(z)$ is supported in S_0. Hence $\sigma(Z) \subset \ell_1^0(S_0)$, a space of dimension $|S_0|$. □

Helemskii's argument for separable P, in the hereditary form needed later.

No complete pieces

Corollary

Let P be topologically projective, $Z \subset P$, and $\|\cdot\|'$ a norm on Z with $(Z, \|\cdot\|')$ complete and $\|\cdot\| \leq K\|\cdot\|'$. Then $\dim Z < \infty$.

Proof.

A separable infinite dimensional $\|\cdot\|'$-closed subspace Z_0 would be $\|\cdot\|$-separable with uncountable dimension. □

Every complete subspace of P is finite dimensional. No infinite dimensional Banach space is topologically projective in **Nor**, although $\ell_1(\Lambda)$ is extremely projective in **Ban**.

Example: completions do not decide

$w_n \geq 1$, $w_n \to \infty$, $P_w = \{x \in \ell_1 : \sum_n w_n|x_n| < \infty\}$ with the ℓ_1-norm: dense in ℓ_1, contains $\ell_1^0(\mathbb{N})$, so $\overline{P_w} = \ell_1$; every complete subspace is finite dimensional (open mapping in $\ell_1(w)$); not topologically projective, as P_w is separable but $\dim P_w = \dim \ell_1(w) > \aleph_0$.

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Theorem (Helemskii 2013; Grønbæk 2016)

P extremely projective $\Rightarrow \overline{P} \cong \ell_1(\Gamma)$ isometrically for some set Γ.

Sketch. By the criterion, $\overline{P}$ is a $(1+\varepsilon)$-retract of the Banach space $\ell_1(B_P)$ for every ε. Then

$$\overline{P}^* \text{ is } (1+\varepsilon)\text{-injective } \forall \varepsilon \;\Rightarrow\; 1\text{-injective} \;\Rightarrow\; \overline{P}^* \cong C(K),\ K \text{ Stonean}$$
$$\Rightarrow\; \overline{P} \cong L_1(\mu) \;\Rightarrow\; \mu \text{ atomic},$$

the last step because $\overline{P}$ has the Schur property as a complemented subspace of $\ell_1(B_P)$ (Grothendieck; Lacey, Thms 3.11.6 and §17).

Two sides of the picture

An extremely projective normed space is, up to isometry,

a **dense subspace of** $\ell_1(\Gamma)$ whose **separable subspaces have countable dimension**.

Question. Is every such subspace extremely projective?
Countable Γ: ✓, and more. Uncountable Γ: ×, but a correction works.

Theorem (back-and-forth)

Let $D \subset \ell_1$ be a dense subspace of countable dimension and $\varepsilon > 0$. There is a linear bijection T of ℓ_1 with

$$\|T\|\|T^{-1}\| \le 1+\varepsilon, \qquad T(D) = \ell_1^0(\mathbb{N}).$$

In particular $d_{BM}(D, \ell_1^0(\mathbb{N})) = 1$.

Plan. (d_j) a Hamel basis of D; $\theta_m \in (0, 1/2]$ with $\prod_m (1+\theta_m)/(1-\theta_m) \le 1+\varepsilon$. Build automorphisms $F_1, F_2, \dots$ of ℓ_1 with $\|F_m - 1\| \le \theta_m$ and put $A_j = F_{2j} F_{2j-1} A_{j-1}$, $T = \lim A_j$.

forth (F_{2j-1}): make $A_{j-1}(d_j)$ finitely supported;

back (F_{2j}): make e_j an element of $A_j(D)$;

while **fixing** a growing finite dimensional $Y_{j-1} \subset \ell_1^0(\mathbb{N})$ that records what was already achieved.

Three lemmas: a *one-vector* lemma, a *hitting* lemma, and a *product* lemma.

The one-vector lemma

Notation: $x|_N = x$ truncated after coordinate N.

Lemma

Let $v \in \ell_1$, $N \geq 0$, $0 < \delta < 1$. There is $R\colon \ell_1 \to \ell_1$, $\|R\| \leq \delta$, with

(i) $R(x) = 0$ *whenever* $x = x|_N$;

(ii) $R(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N})$;

(iii) $(1 - R)(v) \in \ell_1^0(\mathbb{N})$;

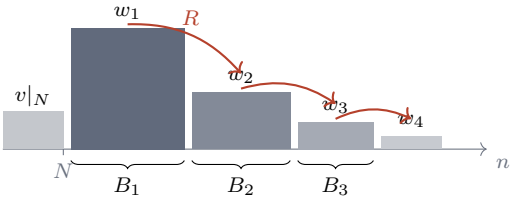
(iv) $(1 - R)^{-1}(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N}) + \mathbb{C}v$.

Hence $T = 1 - R$ is an automorphism of ℓ_1, $\|T\| \leq 1 + \delta$, $\|T^{-1}\| \leq (1 - \delta)^{-1}$, T fixes everything supported in $[1, N]$, $T(\ell_1^0(\mathbb{N})) \subset \ell_1^0(\mathbb{N})$ and

$$T(\ell_1^0(\mathbb{N}) + \mathbb{C}v) = \ell_1^0(\mathbb{N}).$$

One small perturbation of the identity swallows one vector into $\ell_1^0(\mathbb{N})$, without disturbing a prescribed finite block of coordinates and without letting anything else out of $\ell_1^0(\mathbb{N})$ except multiples of v.

Proof: shift along blocks



Cut the tail $v - v|_N$ into finite blocks $w_k = v|_{B_k}$ with

$$\|w_{k+1}\| \le \delta \|w_k\|.$$

Possible since $t_M = \sum_{n>M} |v_n| \downarrow 0$.

$g_k(x) = \sum_{n \in B_k} \overline{\operatorname{sgn} v_n}\, x_n$: then
$g_k(w_j) = \delta_{jk}\|w_k\|$, $|g_k(x)| \le \sum_{B_k} |x_n|$.

$$R(x) = \sum_{k \ge 1} \frac{g_k(x)}{\|w_k\|}\, w_{k+1}, \qquad \|R\| \le \delta.$$

- $R(w_k) = w_{k+1}$: R shifts the blocks one step to the right.
- $R(v) = \sum_k w_{k+1} = v - v|_N - w_1$, so $(1-R)(v) = v|_N + w_1 \in \ell_1^0(\mathbb{N})$.
- For $x \in \ell_1^0(\mathbb{N})$: $R^m(x) = \sum_{k \in K} c_k w_{k+m}$ (finite K), so $(1-R)^{-1}x = \sum_m R^m x = x + \sum_{k \in K} c_k (v - v|_{N_k}) \in \ell_1^0(\mathbb{N}) + \mathbb{C}v$.

Lemma (hitting)

Let $Y \subset \ell_1^0(\mathbb{N})$ be finite dimensional and $e_n \notin Y$. There is $C \geq 1$ such that for every $z \in \ell_1^0(\mathbb{N})$ with $\|z - e_n\| \leq (4C)^{-1}$ there is an automorphism S of ℓ_1 with

$$\|S - 1\| \leq 2C\|z - e_n\| \leq \tfrac{1}{2}, \qquad S|_Y = 1_Y, \qquad S(z) = e_n.$$

Proof. Hahn–Banach: $f \in \ell_1^*$, $f|_Y = 0$, $f(e_n) = 1$, $\|f\| = d^{-1}$, $d = \operatorname{dist}(e_n, Y)$; $C = \max(1, d^{-1})$. Then $|f(z)| \geq 3/4$ and

$$S(x) = x + \frac{f(x)}{f(z)}\,(e_n - z)$$

does the job; it is invertible by the Neumann series. □

A finite dimensional obstacle Y is avoided by a rank-one correction; the closer z is to e_n, the smaller the correction. Density of D will supply z.

The product lemma

Lemma (product)

X Banach, $\theta_m = \|F_m - 1_X\| < 1$, $\sum_m \theta_m < \infty$, $A_j = F_j \cdots F_1$. Then $A_j \to T$ in norm, T is an automorphism, and

$$\|T\| \le \prod_m (1+\theta_m), \qquad \|T^{-1}\| \le \prod_m (1-\theta_m)^{-1}.$$

Completeness is used here, in $\ell_1 = \overline{D}$, never in D.

Choose the budgets $\theta_m \in (0, 1/2]$ in advance with $\prod_m (1+\theta_m)/(1-\theta_m) \le 1+\varepsilon$. Each step may spend at most θ_m on its perturbation; the one-vector lemma ($\|R\| \le \delta$) and the hitting lemma ($\|S-1\| \le 2C\|z - e_n\|$) respect any budget.

The back-and-forth step

Invariants for A_j and the finite dimensional $Y_j \subset \ell_1^0(\mathbb{N})$:

(a) $A_j(d_i) \in Y_j$ for $i \le j$; (forth: basis vectors are caught)

(b) $e_i \in Y_j$ for $i \le j$; (back: unit vectors are caught)

(c) $Y_j \subset A_j(D)$; (everything caught comes from D)

(d) F_{2j-1}, F_{2j} fix Y_{j-1}. (nothing caught is lost)

Forth. $N \ge$ all supports of Y_{j-1}. One-vector lemma for $v = A_{j-1}(d_j)$, N, $\delta = \theta_{2j-1}$: $F_{2j-1} = 1 - R$ fixes Y_{j-1}, and $y_j = F_{2j-1}A_{j-1}(d_j) \in \ell_1^0(\mathbb{N})$. Put $A' = F_{2j-1}A_{j-1}$, $Y' = Y_{j-1} + \mathbb{C}y_j \subset A'(D)$.

Back. If $e_j \in Y'$, done. Otherwise $A'(D)$ is dense, so pick $d \in D$ with $\|A'(d) - e_j\| \le \theta_{2j}/(16C)$.

- one-vector lemma for $A'(d)$ (fixing Y' and e_j), $\delta = \theta_{2j}/2$: $G = 1 - R'$ with $z = GA'(d) \in \ell_1^0(\mathbb{N})$, $\|z - e_j\| \le (4C)^{-1}$;
- hitting lemma: S fixes Y', $S(z) = e_j$, $\|S - 1\| \le 5\theta_{2j}/32$.

$F_{2j} = SG$, $\|F_{2j} - 1\| \le \theta_{2j}$, $Y_j = Y' + \mathbb{C}e_j$, and $e_j = A_j(d) \in A_j(D)$.

Passing to the limit

By the product lemma $A_j \to T$, an automorphism of ℓ_1 with

$$\|T\|\|T^{-1}\| \le \prod_m \frac{1+\theta_m}{1-\theta_m} \le 1+\varepsilon.$$

- $T(D) \subset \ell_1^0(\mathbb{N})$: $A_i(d_i) \in Y_i$ by (a), and all later F_m fix Y_i by (d), so $T(d_i) = A_i(d_i) \in \ell_1^0(\mathbb{N})$.
- $\ell_1^0(\mathbb{N}) \subset T(D)$: by (b), (c) there is $d \in D$ with $A_i(d) = e_i \in Y_i$, and again $T(d) = A_i(d) = e_i$.

So $T(D) = \ell_1^0(\mathbb{N})$. □

What made it work

- each step is a perturbation of the identity of size θ_m, and $\sum \theta_m$ is as small as we like: the constants are *kept*;
- each step is *local*: it fixes a finite block of coordinates and acts on the tail; the one-vector lemma produces nothing outside $\ell_1^0(\mathbb{N}) + \mathbb{C}v$;
- the Hamel basis is countable, so a single ω-sequence of steps exhausts D.

Countable dimension: the characterization

Theorem

Let $\dim P \leq \aleph_0$. *The following are equivalent:*

(i) P *is extremely projective;*

(ii) $\overline{P} \cong \ell_1(\Gamma)$ *isometrically,* Γ *countable;*

(iii) $d_{BM}(P, \ell_1^0(\Gamma)) = 1$ *for some countable* Γ*;*

(iv) *for every* $\varepsilon > 0$ *there is* Λ *with* $d_{BM}(P, \ell_1^0(\Lambda)) < 1 + \varepsilon$.

Proof. (i)⇒(ii): completion theorem; $\overline{P}$ is separable, so Γ is countable. (ii)⇒(iii): $J(P) \subset \ell_1$ is dense of countable dimension; back-and-forth. (iv)⇒(i): a $(1+\varepsilon)$-isomorphism onto $\ell_1^0(\Lambda)$ is a $(1+\varepsilon)$-retraction; criterion. □

Corollary

Every dense subspace of ℓ_1 *of countable dimension is extremely projective and has Banach–Mazur distance* 1 *to* $\ell_1^0(\mathbb{N})$.

No projectivity assumption in (ii): extreme projectivity *is* a property of the completion.

Plan

1. Lifting problems

2. Basic theory

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8. Summary

Extreme points

By the corollary, $\ell_1^0(\mathbb{N}) + V$ is extremely projective for every $V \subset \ell_1$ of countable dimension. Is it ever free?

Lemma

Let $\ell_1^0(S) \subset E \subset \ell_1(S)$ with the ℓ_1-norm. Then

$$\operatorname{ext}(B_E) = \{\lambda e_s : \ s \in S,\ |\lambda| = 1\}, \qquad \operatorname{conv}(\operatorname{ext} B_E) = B_{\ell_1^0(S)}.$$

Proof idea.

The λe_s are extreme in $B_{\ell_1(S)}$, hence in $B_E = B_{\ell_1(S)} \cap E$. If $x \in E$, $\|x\| = 1$, has two non-zero coordinates x_n, x_m, put $t = \min(|x_n|, |x_m|)$ and

$$u = t\big(\operatorname{sgn}(x_n)e_n - \operatorname{sgn}(x_m)e_m\big) \in \ell_1^0(S) \subset E.$$

Then $\|x \pm u\| = (|x_n| \pm t) + (|x_m| \mp t) + \cdots = 1$, so x is not extreme. Finite convex combinations of the λe_s are exactly the elements of $B_{\ell_1^0(S)}$. □

The lemma only sees finitely supported vectors: E may be any space between $\ell_1^0(S)$ and $\ell_1(S)$.

The counterexample

Theorem

Let $\ell_1^0(S) \subset E \subset \ell_1(S)$ with the ℓ_1-norm. If $E \cong \ell_1^0(\Lambda)$ isometrically, then $E = \ell_1^0(S)$.

Proof.

An isometry $T\colon \ell_1^0(\Lambda) \to E$ maps extreme points onto extreme points, so

$$B_E = T\big(\operatorname{conv}\operatorname{ext} B_{\ell_1^0(\Lambda)}\big) = \operatorname{conv}\big(T \operatorname{ext} B_{\ell_1^0(\Lambda)}\big) = \operatorname{conv}\operatorname{ext} B_E = B_{\ell_1^0(S)}.$$

□

Corollary (answer to Helemskii and Grønbæk)

For $V \subset \ell_1$ of countable dimension with $V \not\subset \ell_1^0(\mathbb{N})$, e.g. $V = \mathbb{C}v$, $v = (2^{-n})$, the space $\ell_1^0(\mathbb{N}) + V$ is extremely projective and **not** isometrically isomorphic to any free normed space.

Between $\ell_1^0(\mathbb{N})$ and ℓ_1

space	extremely projective	isometric to a free space
$\ell_1^0(\mathbb{N})$	✓	✓
$\ell_1^0(\mathbb{N}) + V$, $\dim V \leq \aleph_0$, $V \not\subset \ell_1^0(\mathbb{N})$	✓	×
P_w (weighted)	×	×

All three have completion ℓ_1: the difference is invisible on completions. By the theorem, *no* space strictly between $\ell_1^0(\mathbb{N})$ and ℓ_1 is free, yet by the corollary all those of countable dimension are extremely projective.

The next section gives a criterion which distinguishes the projective ones among all dense subspaces, in any dimension.

Plan

1. Lifting problems

2. Basic theory

3. Dimension restrictions

4. Countable dimension

5. Extremely projective spaces which are not free

6. Uncountable dimension

7. Dimension $\aleph_1$

8. Summary

What changes

Countable dimension: dense in ℓ_1 + countable dimension $\Rightarrow$ extremely projective.

Hope. Dense in $\ell_1(\Gamma)$ + all separable subspaces of countable dimension $\Rightarrow$ extremely projective?

No. Already for $\Gamma = \omega_1$ (ladder spaces, Section 7).

What is true (Theorem 6.7)

A dense $P \subset \ell_1(\Gamma)$ is extremely projective $\iff$ Γ has a continuous filtration with **countable blocks** along which the pieces $P \cap \ell_1(\Gamma_\delta)$ are dense, form a **continuous chain**, and have **countable dimensional successive quotients**.

Two tools:

- a transfinite back-and-forth along such a filtration, each countable block solved by the countable theorem, perturbations confined to disjoint blocks;
- closed pairs, an elementary-submodel argument producing such a filtration for every projective P.

Filtrations and blocks

A filtration of Γ: an increasing family $(\Gamma_\delta)_{\delta\le\lambda}$, $\Gamma_0=\varnothing$, $\Gamma_\lambda=\Gamma$, continuous at limits. Blocks $N_\delta=\Gamma_{\delta+1}\setminus\Gamma_\delta$, so $\Gamma_\delta=\bigsqcup_{\delta'<\delta}N_{\delta'}$ and $\ell_1(\Gamma_\delta)=\big(\bigoplus_{\delta'<\delta}\ell_1(N_{\delta'})\big)_{\ell_1}$.

For $P\subset\ell_1(\Gamma)$: $P_\delta=P\cap\ell_1(\Gamma_\delta)$ and

$$D_\delta=\{x|_{N_\delta}:x\in P_{\delta+1}\}\subset\ell_1(N_\delta),\qquad D_\delta\cong P_{\delta+1}/P_\delta.$$

Theorem (transfinite back-and-forth)

Let P be dense in $\ell_1(\Gamma)$ and (Γ_δ) a filtration with countable blocks such that

(i) *P_δ is dense in $\ell_1(\Gamma_\delta)$ for all $\delta\le\lambda$;*

(ii) *$P_\delta=\bigcup_{\delta'<\delta}P_{\delta'}$ for limit $\delta\le\lambda$* *(continuity);*

(iii) *$\dim P_{\delta+1}/P_\delta\le\aleph_0$ for all $\delta<\lambda$* *(small blocks).*

Then for every $\varepsilon>0$ there is an automorphism T of $\ell_1(\Gamma)$ with $T(P)=\ell_1^0(\Gamma)$, $\|T\|\le1+\varepsilon$, $\|T^{-1}\|\le1$.

Each block is solved first: D_δ is dense in $\ell_1(N_\delta)$ of countable dimension, so the countable theorem gives U_δ with $U_\delta(D_\delta)=\ell_1^0(N_\delta)$, $\|U_\delta\|\le1+\theta$, $\|U_\delta^{-1}\|\le1$.

The shape of T

Recursively choose $V_\delta\colon \ell_1(N_\delta) \to \ell_1(\Gamma_\delta)$ with $\|V_\delta\| \le \theta$, and put for $x \in \ell_1(\Gamma_\delta)$

$$W_\delta(x) = x + \sum_{\delta'<\delta} V_{\delta'}(x|_{N_{\delta'}}), \qquad U_{<\delta} = \bigoplus_{\delta'<\delta} U_{\delta'}, \qquad T_\delta = U_{<\delta} W_\delta.$$

- The $N_{\delta'}$ are disjoint and $\|x\| = \sum_{\delta'} \|x|_{N_{\delta'}}\|$, so $\|W_\delta - 1\| \le \theta$ regardless of δ: no product of λ factors ever appears.
- $\|U_{<\delta}\| \le 1+\theta$, $\|U_{<\delta}^{-1}\| \le 1$ (ℓ_1-sum).
- T_δ extends $T_{\delta'}$ for $\delta' < \delta$.

Inductive claim

$$T_\delta(P_\delta) = \ell_1^0(\Gamma_\delta) \qquad \text{for all } \delta \le \lambda.$$

Limit stages are free: by (ii) and compatibility, $T_\mu(P_\mu) = \bigcup_{\delta<\mu} T_\delta(P_\delta) = \bigcup_{\delta<\mu} \ell_1^0(\Gamma_\delta) = \ell_1^0(\Gamma_\mu)$.

Compare with the countable proof: there T was an infinite product of perturbations and the limit stage needed the product lemma. Here the perturbations live on disjoint blocks and *add* instead of multiplying.

The successor step

Let $U = U_\delta$, $y_n = U^{-1}(e_n)$ for $n \in N_\delta$: a Hamel basis of D_δ with $\|\sum c_n y_n\| \geq (1+\theta)^{-1} \sum |c_n|$.

- Lift: $x_n \in P_{\delta+1}$ with $x_n|_{N_\delta} = y_n$.
- Correct: P_δ is dense in $\ell_1(\Gamma_\delta)$ by (i), so there is $z_n \in \ell_1(\Gamma_\delta)$ with

$$x_n|_{\Gamma_\delta} + z_n \in P_\delta, \qquad \|z_n\| \leq \theta(1+\theta)^{-2}.$$

- Define $V_\delta(\sum c_n y_n) = \sum c_n W_\delta(z_n)$ on D_δ; then $\|V_\delta\| \leq \theta$, extend to $\ell_1(N_\delta)$.

Key inclusion

$$x|_{\Gamma_\delta} + W_\delta^{-1} V_\delta(x|_{N_\delta}) \in P_\delta \qquad \text{for every } x \in P_{\delta+1}.$$

($x|_{N_\delta} = \sum c_n y_n$; $x - \sum c_n x_n$ vanishes on N_δ, hence lies in P_δ.)

Then for $x \in P_{\delta+1}$:

$$T_{\delta+1}(x) = T_\delta\big(x|_{\Gamma_\delta} + W_\delta^{-1} V_\delta(x|_{N_\delta})\big) + U(x|_{N_\delta}) \in \ell_1^0(\Gamma_\delta) + \ell_1^0(N_\delta),$$

and $T_{\delta+1}(x_n) \in \ell_1^0(\Gamma_\delta) + e_n$ gives the reverse inclusion. □

Where do good filtrations come from?

Fix a dense topologically projective $P \subset \ell_1(\Gamma)$, a retraction $\sigma\colon P \to \ell_1^0(S)$, $\pi\colon \ell_1^0(S) \to P$, $\pi\sigma = 1_P$, $\|\pi\| \le 1$, and sequences $p_n^\gamma \in P$ with $p_n^\gamma \to e_\gamma$. $P_A = P \cap \ell_1(A)$.

Definition

(A, B), $A \subset \Gamma$, $B \subset S$, is closed if

(a) $\operatorname{supp} \pi(e_s) \subset A$ for $s \in B$;

(b) $\operatorname{supp} \sigma(x) \subset B$ for $x \in P_A$;

(c) $\operatorname{supp} p_n^\gamma \subset A$ for $\gamma \in A$, $n \in \mathbb{N}$.

Lemma

For a closed pair: $P_A = \pi(\ell_1^0(B))$, P_A is dense in $\ell_1(A)$, and $\sigma|_{P_A}$, $\pi|_{\ell_1^0(B)}$ exhibit P_A as a retract of $\ell_1^0(B)$. So P_A is topologically projective.

Proof. (a): $\pi(\ell_1^0(B)) \subset P_A$. (b): $x = \pi\sigma(x) \in \pi(\ell_1^0(B))$. (c): $\overline{P_A} \ni e_\gamma$ for $\gamma \in A$. □

Closed pairs are the "elementary submodels" of the situation.

Countable quotients and countable closing off

Lemma (countable quotient)

(A, B) closed, $A' \supset A$ with $A' \setminus A$ countable. Then $\dim P_{A'}/P_A \leq \aleph_0$.

Proof. $D = \{x|_{A'\setminus A} : x \in P_{A'}\} \cong P_{A'}/P_A$ is separable. The map $\rho(y) = \sigma(x)|_{S\setminus B}$ ($x \in P_{A'}$, $x|_{A'\setminus A} = y$) is well defined, bounded and **injective**: $\rho(y) = 0$ forces $\sigma(x) \in \ell_1^0(B)$, so $x = \pi\sigma(x) \in P_A$ and $y = 0$. A bounded injection into a free space bounds the dimension by the density (the argument of the dimension bound). □

Lemma (countable closing off)

(A, B) closed, $N \subset \Gamma$ countable. There is a closed pair $(A', B') \supset (A \cup N, B)$ with $A' \setminus A$, $B' \setminus B$ countable.

Proof. Alternate ω times: add supports of σ of a basis of P_{A_k} *modulo* P_A (countable by the first lemma), then supports of $\pi(e_s)$ and of the p_n^γ for the new points. Unions. (a), (c) are clear; (b) by density: $\bigcup_k P_{A_k}$ is dense in $\ell_1(A')$, and σ is continuous. □

Subtlety: $\operatorname{supp} x$ need not sit in one A_k, so density, not a support argument, proves (b).

Lemma (unions)

The union of an increasing chain $((A_\delta, B_\delta))_{\delta<\lambda}$ of closed pairs is a closed pair, and

$$P_{A_\lambda} = \pi(\ell_1^0(B_\lambda)) = \bigcup_{\delta<\lambda} \pi(\ell_1^0(B_\delta)) = \bigcup_{\delta<\lambda} P_{A_\delta}.$$

Proof. (a), (c) are inherited. (b): $\bigcup_\delta P_{A_\delta}$ is dense in $\ell_1(A_\lambda)$ (each P_{A_δ} is dense in $\ell_1(A_\delta)$), and the same continuity argument gives $\sigma(x) \in \ell_1^0(B_\lambda)$ for $x \in P_{A_\lambda}$. The formula for P_{A_λ} follows from $P_A = \pi(\ell_1^0(B))$ for closed pairs. □

Continuity (ii) is **automatic** along chains of closed pairs, for any cofinality. This is exactly what the ladder spaces of the next section lack: there is no retraction σ, and the chain jumps on a stationary set.

The criterion

Theorem

Let P be a dense subspace of $\ell_1(\Gamma)$. The following are equivalent:

(i) *P is extremely projective;*

(ii) *P is topologically projective;*

(iii) *Γ has a filtration with countable blocks satisfying (a) dense pieces, (b) continuity, (c) countable successive quotients.*

If so, for every $\varepsilon > 0$ there is an automorphism T of $\ell_1(\Gamma)$ with $T(P) = \ell_1^0(\Gamma)$, $\|T\| \leq 1 + \varepsilon$, $\|T^{-1}\| \leq 1$; in particular $d_{BM}(P, \ell_1^0(\Gamma)) = 1$.

Proof. (ii)⇒(iii): enumerate $\Gamma = \{\gamma_\alpha\}_{\alpha<\kappa}$ and build closed pairs: $A_0 = B_0 = \varnothing$; close off $A_\alpha \cup \{\gamma_\alpha\}$ at successors (countable increment); unions at limits. (a) closed pairs are dense; (b) unions lemma; (c) countable quotient lemma. (iii)⇒(i): the transfinite theorem gives T; then $T|_P$ and $T^{-1}|_{\ell_1^0(\Gamma)}$ make P a $(1+\varepsilon)$-retract of $\ell_1^0(\Gamma)$; the criterion of Section 2. □

One transfinite construction with countable steps; no induction on cardinals.

The characterization in all dimensions

Theorem

For a normed space P the following are equivalent:

(i) *P is extremely projective;*

(ii) *P is topologically projective and $\overline{P} \cong \ell_1(\Gamma)$ isometrically;*

(iii) *$d_{BM}(P, \ell_1^0(\Gamma)) = 1$ for some set Γ;*

(iv) *for every $\varepsilon > 0$ there is Λ with $d_{BM}(P, \ell_1^0(\Lambda)) < 1 + \varepsilon$.*

In (iii), $|\Gamma| = \dim P$ for infinite dimensional P, and the distance is realized by bijections $T\colon \overline{P} \to \ell_1(\Gamma)$ with $T(P) = \ell_1^0(\Gamma)$ and $\|T\|\|T^{-1}\|$ arbitrarily close to 1.

Proof. (i)⇒(ii): completion theorem. (ii)⇒(iii): $J(P)$ is a dense topologically projective subspace of $\ell_1(\Gamma)$; general theorem. (iv)⇒(i): criterion. □

Grønbæk's conjecture

Every extremely projective normed space is **almost isometric** to $\ell_1^0(\Gamma)$, $|\Gamma| = \dim P$. The conjecture holds in the almost isometric form and fails in the isometric form.

Helemskii's question on dense subspaces

Which dense subspaces $P \subset \ell_1(\Lambda)$ are extremely projective?

Answer

P extremely projective $\iff$ P topologically projective $\iff$ P admits a filtration with countable blocks, dense pieces, continuity and countable quotients.

- Λ countable: $\iff \dim P \leq \aleph_0$.
- Λ uncountable: a genuine restriction beyond dimension; the continuity condition fails for the ladder spaces of the next section.
- The answer refers only to the subspaces $P \cap \ell_1(A)$, and the coordinates of $\ell_1(\Lambda)$ are intrinsic: the extreme points of the unit ball are the λe_γ (Section 5). No retraction onto a free space is mentioned.
- Topological and extreme projectivity coincide for dense subspaces of $\ell_1(\Gamma)$. This is a feature of the *exact* norm: if E is $\ell_1(\Gamma)$ with an equivalent, not almost isometric, norm, then $J(\ell_1^0(\Gamma)) \subset E$ is topologically but not extremely projective.

Plan

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3. Dimension restrictions

4. Countable dimension

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6. Uncountable dimension

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8. Summary

Set theory in one slide

- ω_1 = the first uncountable ordinal; its elements are the countable ordinals. Every $x \in \ell_1(\omega_1)$ has countable support, hence lives in some $\ell_1(\delta)$, $\delta < \omega_1$.
- $C \subset \omega_1$ is closed unbounded (club) if unbounded and closed under countable suprema. The limit ordinals form a club; two clubs meet in a club; for $h\colon \omega_1 \to \omega_1$ the ordinals closed under h form a club.
- $E \subset \omega_1$ is stationary if it meets every club; *non-stationary* = disjoint from some club = "small".
- **Solovay.** ω_1 is the union of $\aleph_1$ pairwise disjoint stationary sets.

The objects

P dense in $\ell_1(\omega_1)$, $P_\delta = P \cap \ell_1(\delta)$ for $\delta < \omega_1$: an increasing ω_1-chain of subspaces with union P.

All that matters about an ω_1-chain is what happens on a club. The enumeration of the index set changes the chain only off a club.

The invariant and the criterion

Definition

$\Delta(P) = \{\delta < \omega_1 \text{ limit} : \ P_\delta \neq \bigcup_{\delta' < \delta} P_{\delta'}\}$, the set of jumps of the chain.

Theorem

A dense $P \subset \ell_1(\omega_1)$ is extremely projective $\iff$

(i) *every P_δ, $\delta < \omega_1$, has countable dimension (equivalently: every separable subspace of P does), and*

(ii) $\Delta(P)$ *is not stationary.*

Proof. ($\Rightarrow$) (i) is the dimension bound. For (ii) take the chain of closed pairs (A_α, B_α) from the criterion with $\gamma_\alpha = \alpha$. On a club, $A_\delta = \delta$; there $P_\delta = P_{A_\delta} = \bigcup_{\alpha<\delta} P_{A_\alpha}$ by the unions lemma, and each A_α sits below some $\delta' < \delta$.
($\Leftarrow$) Density of P_δ holds on a club C_1; a club C_2 misses $\Delta(P)$. Enumerate $C_1 \cap C_2 \cup \{0\}$ as (c_α) and take $\Gamma_\alpha = c_\alpha$: countable blocks, (a) by C_1, (b) by C_2 and continuity of the enumeration, (c) by (i). Apply the criterion. □

Both conditions are needed; the invariant is invariant

- $P_w + \ell_1^0(\omega_1 \setminus \omega)$ (P_w the weighted space on ω): $\Delta(P) = \{\omega\}$, not stationary, but $P_\omega = P_w$ has uncountable dimension. Not projective.
- Ladder spaces on stationary sets (next slide): (i) holds, (ii) fails.

Proposition

If T is a bounded bijection of $\ell_1(\omega_1)$ with $T(P) = P'$, then $\Delta(P) \triangle \Delta(P')$ is not stationary. Hence the class of $\Delta(P)$ modulo non-stationary sets is an isomorphism invariant of the normed space P, independent of the enumeration of the index set.

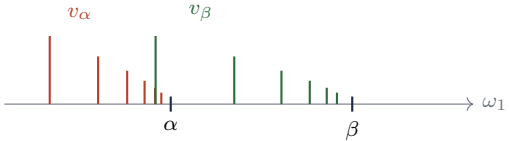
Proof. $F(\delta) = \sup_{\beta<\delta}(\sup \operatorname{supp} Te_\beta + 1)$, similarly G for T^{-1}. For limit δ closed under F and G (a club): $T(\ell_1(\delta)) = \ell_1(\delta)$, so $T(P_\delta) = P'_\delta$ and $T\big(\bigcup_{\delta'<\delta} P_{\delta'}\big) = \bigcup_{\delta'<\delta} P'_{\delta'}$; jumps correspond. An isomorphism $P \to P'$ extends to the completions. □

Ladder spaces

E a set of limit ordinals $< \omega_1$. For $\alpha \in E$ fix a ladder $\alpha_1 < \alpha_2 < \cdots \nearrow \alpha$ and put

$$v_\alpha = \sum_{n \geq 1} 2^{-n} e_{\alpha_n},$$

$$P_E = \ell_1^0(\omega_1) + \operatorname{span}\{v_\alpha : \alpha \in E\}.$$



Two ladders share only finitely many rungs.

Theorem

(i) $(P_E)_\delta = \ell_1^0(\delta) + \operatorname{span}\{v_\alpha : \alpha \in E,\ \alpha \leq \delta\}$: *countable dimension, dense, hence* ***extremely projective*** *for every* $\delta < \omega_1$;

(ii) $\Delta(P_E) = E$ (v_δ *enters at stage* δ *but is not a finite combination of earlier vectors modulo* $\ell_1^0(\delta)$);

(iii) P_E *is extremely projective* $\iff$ E *is not stationary;*

(iv) $E \triangle E'$ *stationary* $\Rightarrow P_E \ncong P_{E'}$.

$2^{\aleph_1}$ spaces, and an old theory in new clothes

Solovay: disjoint stationary $S_\xi \subset \omega_1$, $\xi < \omega_1$. For $X \subset \omega_1$ put $E_X = \bigcup_{\xi \in X} S_\xi$. Then $E_X \triangle E_Y \supset S_\xi$ for $X \neq Y$.

$2^{\aleph_1}$ pairwise non-isomorphic dense subspaces of $\ell_1(\omega_1)$, all with extremely projective countable pieces; only $P_\varnothing = \ell_1^0(\omega_1)$ is extremely projective. All of them contain the same dense subspace $\ell_1^0(\omega_1)$.

almost free abelian groups	extremely projective normed spaces
Pontryagin, Hill: filtrations by countable subgroups with free quotients	Theorem 6.7: filtrations with countable blocks and countable quotients
Eklof's invariant $\Gamma(G)$	$\Delta(P)$ modulo non-stationary sets
$\aleph_1$-free non-free groups built from ladder systems	the spaces P_E

In the group case the picture above $\aleph_1$ depends on set theory; expect the same here.

Plan

1. Lifting problems
2. Basic theory
3. Dimension restrictions
4. Countable dimension
5. Extremely projective spaces which are not free
6. Uncountable dimension
7. Dimension $\aleph_1$
8. Summary

Where each ingredient was consumed

ingredient	used for	where
universal property of $\ell_1^0(\Lambda)$	free spaces are extremely projective; canonical morphism π_P	§2
retraction σ onto a free space	supports are finite: dimension bound, closed pairs, countable quotients	§3, §6
Grothendieck (via duality of $\overline{P}$)	completion is $\ell_1(\Gamma)$	§4
exact ℓ_1-norm: disjoint blocks add	$\|R\| \leq \delta$ in the one-vector lemma; $\|W_\delta - 1\| \leq \theta$ transfinitely	§4, §6
Hahn–Banach, Neumann series in $\overline{P}$	hitting lemma; invertibility	§4
density of P_δ in $\ell_1(\Gamma_\delta)$	corrections z_n at successor steps	§6
extreme points of B_{ℓ_1}	no space strictly between $\ell_1^0(S)$ and $\ell_1(S)$ is free; coordinates are intrinsic	§5, §7
clubs and stationary sets, Solovay	the $\aleph_1$ criterion; $2^{\aleph_1}$ ladder spaces	§7

Comparison with Grønbæk's proof

Grønbæk: topologically projective $\Rightarrow$ isomorphic to $\ell_1^0(\Lambda)$, in two steps.

1. Transfinite construction: every subspace $Y \subset \ell_1^0(\Lambda)$ contains a complemented copy of $\ell_1^0(\Lambda')$, $|\Lambda'| = \operatorname{dens} Y$.
   Inspection: the copy is $(1+\varepsilon)$-isomorphic and $(1+\varepsilon)$-complemented.
2. Pełczyński's decomposition method.
   Here the constants are lost.

- decomposition method: $Y \oplus \ell_1^0(\Lambda) \cong \ell_1^0(\Lambda)$, then $Y \cong Y \oplus \ell_1^0(\Lambda) \cong \ell_1^0(\Lambda)$, with uncontrolled constants;
- back-and-forth: an automorphism of $\ell_1(\Gamma)$ of distortion $< 1+\varepsilon$ moving P exactly onto $\ell_1^0(\Gamma)$.

The price of keeping the constants is the continuity condition (ii), and the closed pairs show that it is always available for projective spaces.

Open problems

- Dimension $\geq \aleph_2$. Can the filtration condition be replaced by a condition on the pieces $P \cap \ell_1(A)$ for A in a club of subsets of Γ? Is there a dense subspace of $\ell_1(\omega_2)$ which is not projective although every piece of size $\aleph_1$ is? For groups the answers depend on set theory.
- Dense subspaces of other Banach spaces: for $E \ncong \ell_1(\Gamma)$ almost isometrically there are topologically but not extremely projective dense subspaces. Which E admit extremely projective dense subspaces at all? (Only those with $E \cong \ell_1(\Gamma)$ isometrically.)
- Extreme projectivity over non-zero algebras: Helemskii's original setting of sequence algebras. Is there an analogue of the almost isometric characterization, of the counterexample, of the invariant?

The real question

In **Nor** the free objects are tiny and the projective objects are almost free. What replaces the Banach space toolbox when every complete subspace is finite dimensional?

Summary

- **Extremely projective** $= C$-topologically projective for every $C > 1$, in the category of normed spaces.
- **Counterexample**: $\ell_1^0(\mathbb{N}) + \mathbb{C}v$ is extremely projective and not free. The isometric conjecture of Helemskii and Grønbæk fails.
- **Characterization**: extremely projective $\iff$ topologically projective with completion $\ell_1(\Gamma) \iff d_{BM}(P, \ell_1^0(\Gamma)) = 1$. The conjecture holds almost isometrically.
- **Countable dimension**: a property of the completion alone.
- **Uncountable dimension**: filtrations with countable blocks; for $\aleph_1$ a stationary-set invariant and $2^{\aleph_1}$ ladder spaces.
- **Method**: back-and-forth, countable and transfinite, with perturbations on disjoint blocks; closed pairs.

Thank you!

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Backup: no 1-topologically projective spaces

Let $X = c_{00}$ with the sup-norm, $s = \sum_n 2^{-n}(1-2^{-n})$ and

$$f(x) = s^{-1} \sum_{n \geq 1} 2^{-n}(1-2^{-n})\, x_n, \qquad \|f\| \leq 1.$$

- f is coisometric: for $|z| < 1$ pick N with $|z| < s^{-1} \sum_{n \leq N} 2^{-n}(1-2^{-n})$ and $x = t \operatorname{sgn}(z) \sum_{n \leq N} e_n$ with suitable $t \in [0,1)$; then $f(x) = z$, $\|x\| < 1$.
- f does not attain its norm: $|f(x)| = 1$, $\|x\| \leq 1$ would force $x_n = \operatorname{sgn} f(x)$ for all n.

Let $P \neq 0$ be 1-topologically projective, $p \in P$, $\|p\| = 1$, and $g \in P^*$ with $\|g\| = g(p) = 1$ (Hahn–Banach). Lift g along f: $\psi\colon P \to X$, $f\psi = g$, $\|\psi\| \leq 1$. Then $x = \psi(p) \in B_X$ and $f(x) = g(p) = 1$. Contradiction. □

The same functional on ℓ_1, $x \mapsto \sum (1-2^{-n}) x_n$, shows that with the open-ball convention 1-topological projectivity is empty in **Ban** too. With closed balls (strictly coisometric operators) one recovers the metric theory; for extreme projectivity the choice is immaterial.

Backup: the product lemma

$\theta_m = \|F_m - 1_X\| < 1$, $\sum \theta_m < \infty$, $A_j = F_j \cdots F_1$, $M = \prod(1+\theta_m)$, $M' = \prod(1-\theta_m)^{-1}$.

- $\|A_j\| \le M$ and, by the Neumann series, $\|F_m^{-1}\| \le (1-\theta_m)^{-1}$, so $\|A_j^{-1}\| \le M'$.
- $\|A_j - A_{j-1}\| = \|(F_j - 1)A_{j-1}\| \le \theta_j M$: Cauchy, $A_j \to T$, $\|T\| \le M$.
- $\|A_j^{-1} - A_k^{-1}\| = \|A_j^{-1}(A_k - A_j)A_k^{-1}\| \le {M'}^2 \|A_k - A_j\|$: Cauchy, $A_j^{-1} \to B$, $\|B\| \le M'$.
- $A_j A_j^{-1} = A_j^{-1} A_j = 1$ in the limit: $TB = BT = 1$. □

In the countable back-and-forth the θ_m are chosen in advance with $\prod(1+\theta_m)/(1-\theta_m) \le 1+\varepsilon$; every step then has a fixed budget θ_m for its perturbation, and the budget is respected by the one-vector lemma ($\|R\| \le \delta$) and the hitting lemma ($\|S - 1\| \le 2C\|z - e_n\|$, with z chosen close enough to e_n by density).